Computer Programming In QBasic

QBASIC COMMANDS

1. ***CLS***: This command is used to clear the screen.

2. ***PRINT***: Print command is used to display the output on the screen.
E.g. **Print "HELLO WORLD!!!"**
Print 80 * 8
Print – Only Print command will leave blank space.
Print Tab(10) "Navrachana" – will print Navrachana on 10 column.

3. ***REM***: It stands for Remark. It gives an explanation of the program or of the statements in the program thereby making the program more understandable to the reader. The computer does not execute this statement since whatever is written after REM is ignored by the compiler. REM can be used anywhere and many times in a program.

4. ***LET***: It assigns a value to a variable in a program. It stores a value in the memory location.

SYNTAX:

Let <Variable> = < Constant / Variable or Expression >

e.g.
Let A = 15…….. Assigning constant to a variable
Let A = B………Assigning variable to a variable
Let C = A + B…. Assigning an expression to a variable

Using Let with Numeric Variables:

Let A = 50, *here **A** is a Numeric Variable and '50'* is a Numeric Constant and value '50' is assigned to A and stored in the computer's memory for further calculations.

Using Let with Alphanumeric Variable:
Let N$ = "QBasic Program", here N$ is an Alphanumeric Variable and "QBasic Program" is the Alphanumeric Constant assigned to N$.

NOTE: A numeric data should be assigned to a Numeric variable and an alphanumeric data to an alphanumeric variable otherwise "TYPE MISMATCH" error is displayed.

5. ***END***: This command is usually given at the end of the program. Statements written after end are not executed since the program terminates execution on reading this command.

6. ***INPUT***: This statement allows the user to enter a value for the variable while running the program. A question mark (?) appears on the output screen waiting for the user to enter a relevant data and then press enter key. Once the Return key or Enter key is pressed the data is stored in the variable.

SYNTAX :
INPUT < VARIABLE >
e.g.
Input A……Enter Numeric constant
Input N$…..Enter Alphanumeric constant
Input “Enter name:”;N$….Giving relevant message to avoid erroneous data input

QBASIC REMINDER

A QBASIC program consists of lines containing
1. A line number
2. A QBASIC keyword like PRINT,END etc
3. Each program line begins with positive number.
4. No two lines should have same number.

Chapter I - QBasic Commands

When you open QBasic, you see a blue screen where you can type your program. Let’s begin with the QBasic commands that are important in any program.

PRINT

Command PRINT displays text or numbers on the screen.
The program line looks like this:
PRINT “My name is Nick.”

Type the bolded text into QBasic and press F5 to run the program. On the screen you’ll see

My name is Nick.

Note: you must put the text in quotes, like this – “text”.

The text in quotes is called a string.
If you put the PRINT alone, without any text, it will just put an empty line.

PRINT can also put numbers on the screen.
PRINT 57 will show the number 57. This command is useful for displaying the result of mathematical calculations. But for calculations, as well as for other things in the program, you need to use variables.

Illustration

```
1 CLS      ' this command clears the screen, so it’s
empty
 INPUT "Enter the first number "; a
 INPUT "Enter the second number "; b
 IF b = 0 THEN          '  checks if the second number is
zero, because you can’t divide by zero
    PRINT "the second number cannot be 0.  Try again."
    DO: LOOP WHILE INKEY$ = ""   '  waits for you to
press a key to continue
    GOTO 1 then sends you back to line 1
 END IF
END
```

Note: Do NOT use the Close box to close the editor. You should always go to *FILE* then *EXIT*. If the editor is not closed correctly or if you have not saved your file, you could loose all your data.

QBasic Tutorial

INTRODUCTION TO QBASIC

BASIC stands for **B**eginner's **A**ll **P**urpose **S**ymbolic **I**nstruction **C**ode. It was invented in 1963, at Dartmouth College, by the mathematicians John George Kemeny and Tom Kurtzas.

BASIC is an interpreter which means it reads every line , translates it and lets the computer execute it before reading another. Each instruction starts with a line number.

FEATURES OF QBASIC

1. It is a user friendly language.
2. It is widely known and accepted programming language.
3. It is one of the most flexible languages, as modification can easily be done in already existing
4. program.

5. Language is easy since the variables can be named easily and uses simple English phrases
6. with mathematical expressions.

RULES OF QBASIC

Every programming language has a set of rules that have to be followed while writing a program, following are some rules of QBASIC language:

1. All QBasic programs are made up of series of statements, which are executed in the order in which they are written.

2. Every statement should have at least one QBasic command word. The words that BASIC recognizes are called keywords.

3. All the command words have to be written using some standard rules, which are called "Syntax Rules". Syntax is the grammar of writing the statement in a language. Syntax Errors are generated when improper syntax is detected.

QBASIC DATA

Data is a collection of facts and figures that is entered into the computer through the keyboard. Data is of two types:

1. ***CONSTANT***: Data whose value does not change or remains fixed. There are two types of constants:

(a) NUMERIC CONSTANT: Numbers - negative or positive used for mathematical calculations
e.g. –10, 20, 0

(b) ALPHANUMERIC CONSTANT / STRING: Numbers or alphabets written within double quotes(inverted commas “ “).
e.g. “Computer”, “Operating System”

2. ***VARIABLE***: Data whose value is not constant and may change due to some calculation during the

program execution. It is a location in the computer's memory, which stores the values. Depending on what value is held, Variables are of two types:

(a) NUMERIC VARIABLE: The variable that holds a Numeric Constant for arithmetic calculations (+, - ,*, /) is called a Numeric Variable.
e.g. A = 50, here A is the Numeric Variable
(b) ALPHANUMERIC VARIABLE: The variable that holds an Alphanumeric Constant, which cannot be used for arithmetic calculations, is called Alphanumeric Variable or String Variable. An Alphanumeric variable must end with a $ sign and the Alphanumeric constant must be enclosed in inverted commas.
e.g. Name$ = "Akanksha", here Name$ is an Alphanumeric Variable

TYPES OF MODE IN QBASIC

Once QBASIC program is loaded into the computer memory, it displays Ok prompt. Ok means it ready to accept the commands. QBASIC can be made to translate your instructions in two modes:

- Direct Mode
- Program Mode

1. Direct Mode: The accepts single line instructions from the user and the output is viewed as soon as enter key is pressed. The instructions are not stored in the memory. This mode can be used to do quick calculation. They do not have line numbers. E.g.

Print 3+4

Print "This is the Direct mode in QBasic"

2. Program Mode: The mode is used to type a program which is stored in the memory. They have line numbers. We have to give the command to get the output.
e.g.

```
10 Print 3+4
20 End
RUN
```

Programs are built up with set of instructions or commands. Every programming language has its own SYNTAX (rules) and COMMANDS.

INPUT

INPUT is a command that allows you or anybody else who runs the program to enter the information (text or number) when the program is already running. This command waits for the user to enter the information and then assigns this information to a variable. Since there are two types of variables, the INPUT command may look like this – **INPUT a** (*for a number*), or **INPUT a$** (*for a string*).

Example (Type this program into QBasic and run it by pressing F5)

```
PRINT “What is your name?”
INPUT name$
PRINT “Hi, ”; name$; “, nice to see you!”
PRINT “How old are you?”
INPUT age
PRINT “So you are ”; age; “ years old!”
END
```

<u>Note:</u> The **END** command tells QBasic that the program ends here.
You don’t have to use **PRINT** to ask the user to enter the information. Instead, you can use

INPUT “Enter your name”; name$

and the result will be the same.

<u>GOTO</u>

Quite often you don’t want the program to run exactly in the order you put the lines, from the first to the last. Sometimes you want the program to jump to a particular line. For example, your program asks the user to guess a particular number:

~ ~ ~ ~ ‘some of the program here
INPUT “Guess the number”; n
~ ~ ~ ~ ‘some of the program there

The program then checks if the entered number is correct. But if the user gives the wrong answer, you may want to let him try again. So you use the command **GOTO**, which moves the program back to the line where the question is asked. But first, to show QBasic where to go, you must “label” that line with a number:

1 INPUT “Guess the number”; n ‘this line is labelled with number 1

Then, when you want the program to return to that line, you type

GOTO 1

You can use **GOTO** to jump not only back but also forward, to any line you want. Always remember to label that line. You can have more than one label, but in that case they should be different.

Chapter II
Mathematical Calculations

QBasic was obviously created for us to have fun, play games, draw nice graphics and even make sounds.

But, as you might guess, nothing good comes without a bit of effort that has to be put in it. In the most QBasic programs a bit of math has to be done.
The math… Doh!
If you hate mathematics, don't worry. QBasic will do it all for you, you just need to know how to tell QBasic to do that.

QBasic can perform the following mathematical operations: